

Steven Wilcox

Machine Learning Scientist



swilcox.dev



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ABOUT

- Physicist and mathematician turned data scientist with over 8 years of work experience in data science, enterprise machine learning engineering, and product development.

SKILLS

- **Languages:** Python, Javascript, MATLAB, $\text{\LaTeX}$
- **Machine Learning:** Deployment, Testing, Numpy, Pandas, SciPy, Scikit-Learn, Keras, PyTorch, Tensorflow, LLM APIs (e.g. OpenAI, Claude), RAG, LLM Prompting, LLM Fine-Tuning
- **Data:** SQL/NoSQL (PostgreSQL, MongoDB), ORMs/ODMs (SQLAlchemy, Mongoose), vector databases
- **DevOps:** Bash, Git/GitHub/GitLab, CircleCi (CI/CD), Docker, Google Cloud Platform
- **Software Development:** Object-Oriented Programming (OOP), Algorithms, Linting (PyLint, EsLint), Testing (PyTest), Documentation (Pdoc)
- **Visualization:** Matplotlib, Seaborn, Plotly, Bokeh

RELEVANT WORK EXPERIENCE

SIRL | Founder, Data Scientist *November 2024 – Present* | Columbus, OH

- SIRL is a mixed-reality role-playing game (RPG) for personal development, allowing players to realize their aspirational self in real life (IRL) through a series of in-game actions and similarly aligned real-world challenges.
- We are developing B-CODE: a cognitive behavioral programming language, which interfaces between a digital representation of the player's personality and an LLM powered side-kick for the sake of planning character development.

FREELANCE CONSULTING | Data Scientist *January 2024 - November 2024* | Columbus, OH

- Developed models utilizing nonlinear signal processing and interpretable machine learning for embedded systems.

OXYML | Product Lead *January 2023 – December 2023* | Remote

- Initiated and developed consulting relationships with asset managers and financial institutions 10Bn-200Bn+ assets under management (AUM) for the purpose of informing product development and accelerating AI adoption in asset management workflows.
- Socialized OxyML's compliance, search, and generative portfolio products. Recruited feedback and crafted proposals for enterprise AI pipelines individually tailored to the needs of prospective clients in Columbus, OH.
- This work directly resulted in the development of two completely novel AI asset management products, which I helped design and showcase.

CARDINAL HEALTH | Sr. Data Scientist and ML Engineer *March 2021 – July 2022* | Dublin, OH

- Lead data scientist for EdgePark document understanding initiative, aimed at simplifying patient UX flows.
- Conducted a comprehensive literature review for deep learning approaches to structured document understanding tasks.
- Designed custom synthetic data generation, training, inference, and model maintenance workflows for extracting and classifying data from structured documents utilizing classical edge detection methods, U-Net segmentation, and Google's AutoML object detection services.
- Lead the productionization of an enterprise machine learning model for data quality.
- Developed a comprehensive MLOps Center of Excellence with accompanying multimedia content, tutorials, and code templates. Implemented transcription and named entity recognition (NER) service using websockets and synthetic call transcript data for the development of a real-time data analytics customer service console.
- Helped design Cardinal Health's internal MLOps platform, standards, and workflows utilizing Google Cloud Platform's Vertex AI services.

THE COVID TUTORING INITIATIVE | Full-Stack Web Developer *January 2020 – Aug 2020 | Palo Alto, CA*

- Managed and collaborated with a small developer, designer, and intern team to build an online tutoring platform prototype (ReactJS, Firebase) to increase educational access to students affected by Covid-19 school closures.
- Volunteered my time to support students academically and sponsor high school and college developer internships.

AJ TUTORING | Private Tutor *August 2017 – Nov 2019 | Menlo Park, CA*

- Over 2000 hours, 100 students, and 15 courses & tests.
- Coached students to achieve SAT score increases of 200+ points.

UNIVERSITY OF FLORIDA LANDMINE DETECTION LAB | Data Scientist *May 2015 – Aug 2016 | Gainesville, FL*

- Implemented deep learning algorithms for landmine detection using ground-penetrating radar (US Army grant).
- Introduced a new 'research pipeline' using **GPU accelerated deep learning in Python with Keras and Theano**.

UF PHYSICS | Physics Instructor and TA *August 2013 – May 2015 | Menlo Park, CA*

- Taught over 300 students in labs & courses.
- Graded assignments, designed and proctored quizzes.
- Held weekly office hours.

THE LARGE HADRON COLLIDER AT CERN | Graduate Researcher and Data Analyst *May 2014 – Aug 2014 | Meyrin, CH*

- Collaborated with the CMS group at the Large Hadron Collider to understand the aging process of the cathode strip chambers in the Endcap Muon System of the CMS detector.
- Presented my analysis on the viability of temperature sensors as a diagnostic for detector aging to the CMS group.

UNIVERSITY OF WASHINGTON | GRADUATE RESEARCH TRAINING GROUP *| July - August 2012 | Seattle, Washington*

- **The Radon and the X-Ray Transform** - A mathematical introduction to the essential properties of these transforms and their implementation (MATLAB).
- **A Mathematical Approach to Cancer Growth** - A mini-course including different models for solid tumor growth (dynamical systems).

EDUCATION

UNIVERSITY OF FLORIDA

M.S. Physics

GAINESVILLE, FL

Aug 2013 - Aug 2015

NEW COLLEGE OF FLORIDA

B.A. Mathematics and Physics (Honors)

SARASOTA, FL

Aug 2009 - May 2013

PROFESSIONAL DEVELOPMENT

- **Fundamentals:** CS229 Machine Learning (Stanford)
- **Vision:** CS231N: Convolutional Neural Networks for Visual Recognition (Stanford)
- **NLP:** CS224N: Natural Language Processing with Deep Learning (Stanford)
- **Additional Graduate Courses:** Machine Learning; Statistics; Partial Differential Equations; Quantum, Statistical, and Classical Mechanics; Electromagnetism, Probability and Statistics
- **AI Conferences and Communities:** TWIMLcon 2021, NeurIPS 2020, The ML Collective, Fast.ai
- **MOOCs:**
 - Testing and Monitoring of Machine Learning Model Deployments
 - Deployment of Machine Learning Models
 - Feature Selection for Machine Learning
 - The Complete Node.js Developer Course (3rd Edition)
 - The Complete React Developer Course (w/ Hooks and Redux)

ACADEMIC HIGHLIGHTS

- **Publication** Investigation of the switching wave propagation in linear chains of magnetic elements (Journal of Applied Physics)
- **Awarded** 3 undergraduate research internships and a graduate fellowship to work at the Large Hadron Collider at CERN.
- **Ranked** in the top 14th percentile in the nation on the Putnam Exam, "the world's hardest math exam".